

# Lecture Notes · Arithmetic · Session 1 — Divisibility, Euclidean division and the Euclidean algorithm

All the session's material with complete proofs: division with remainder (existence via well-ordering, and uniqueness), the lemma that justifies the Euclidean algorithm, and Bézout's identity by its two routes — the structural one and the extended algorithm with a fully worked example. Exercises and a Going further section at the end.

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## 1.1 Divisibility

**Definition 1.1.** For  $a, b \in \mathbb{Z}$ :  $a$  **divides**  $b$ , written  $a \mid b$ , if there exists  $c \in \mathbb{Z}$  with  $b = ac$ .

**Proposition 1.2.** For arbitrary integers: 1. (*Transitivity*)  $a \mid b$  and  $b \mid c \Rightarrow a \mid c$ . 2. (*Combinations*) if  $a \mid b$  and  $a \mid c$ , then  $a \mid (xb + yc)$  for all  $x, y \in \mathbb{Z}$ . 3. If  $a \mid b$  and  $b \neq 0$ , then  $|a| \leq |b|$ .

*Proof.* (1)  $b = au, c = bv \Rightarrow c = a(uv)$ . (2)  $b = au, c = av \Rightarrow xb + yc = a(xu + yv)$ . (3)  $b = ac$  with  $c \neq 0$ , so  $|c| \geq 1$  and  $|b| = |a||c| \geq |a|$ . ■

Property (2), so modest, is the key to half the session: Bézout lives off it.

## 1.2 Euclidean division

**Theorem 1.3 (division with remainder).** For  $a \in \mathbb{Z}$  and  $b > 0$  there exist **unique** integers  $q$  (quotient) and  $r$  (remainder) with

$$a = bq + r, \quad 0 \leq r < b.$$

*Proof. Existence.* The set  $S = \{a - bq : q \in \mathbb{Z}, a - bq \geq 0\}$  is nonempty (for  $q$  very negative,  $a - bq$  becomes positive). By the **well-ordering principle** (M0: every nonempty subset of  $\mathbb{N}$  has a least element),  $S$  has a least element  $r = a - bq_0 \geq 0$ . We claim  $r < b$ : if we had  $r \geq b$ , then  $r - b = a - b(q_0 + 1) \geq 0$  would be in  $S$  and smaller than  $r$ , against minimality.

**Uniqueness** (the “subtract and bound” trick). If  $a = bq + r = bq' + r'$  with  $0 \leq r, r' < b$ , subtracting:  $b(q - q') = r' - r$ . The right-hand side satisfies  $|r' - r| < b$ ; the left-hand side is a multiple of  $b$ . The only multiple of  $b$  with absolute value less than  $b$  is 0: hence  $r = r'$  and, since  $b \neq 0$ ,  $q = q'$ . ■

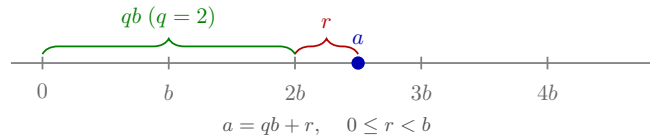


Figure 1: Euclidean division on the number line:  $a$  falls between two consecutive multiples of  $b$ ; the quotient  $q$  counts them and the remainder  $r$  is what is left over ( $0 \leq r < b$ ).

**Remark 1.4 (the convention matters).** For  $a = -7$ ,  $b = 3$ :  $-7 = 3 \cdot (-3) + 2$  — the remainder is 2, not  $-1$ . The convention  $0 \leq r < b$  makes the expression unique and is the one congruences use (Session 3).

*Structural lens (a mention):*  $\mathbb{Z}$  is the model of a world “with division with remainder”. This single property generates everything that follows — and in the Polynomials module we will see that  $K[x]$  shares it, with the **degree** in the role of the absolute value.

### 1.3 Greatest common divisor and the Euclidean algorithm

**Definition 1.5.** For  $a, b$  not both zero,  $\gcd(a, b)$  is the largest of the common divisors. (It exists: the common divisors are bounded by 1.2.3 and the set is nonempty — it contains 1.)

**On “the largest” and the sign.** Divisors come in pairs  $\pm d$ , so the largest in the usual order of  $\mathbb{Z}$  is the **positive** one: thus the  $\gcd$  is an integer  $\geq 0$ , well defined and **unique** — “the largest” is literal. The characterization by divisibility ( $d$  such that every common divisor divides  $d$ ; exercise 8) determines it only **up to sign**, and the convention fixes  $\gcd \geq 0$ . (The case  $a = b = 0$  is excluded on purpose: **every** integer divides 0, so there is no greatest common divisor.) (*In  $K[x]$  — the Polynomials module — the  $\gcd$  is unique up to a nonzero constant factor, and it is normalized by making it monic: the same “up to a unit” issue.*)

**Lemma 1.6 (why the Euclidean algorithm works).** If  $a = bq + r$ , then the **common divisors of  $(a, b)$  are exactly the common divisors of  $(b, r)$** . In particular,  $\gcd(a, b) = \gcd(b, r)$ .

*Proof.* If  $d \mid a$  and  $d \mid b$ : since  $r = a - bq$  is a combination of  $a$  and  $b$ ,  $d \mid r$  (Prop. 1.2.2) — a common divisor of  $(b, r)$ . Conversely, if  $d \mid b$  and  $d \mid r$ :  $a = bq + r$  is a combination of  $b$  and  $r$ , so  $d \mid a$ . The two pairs have *the same set* of common divisors; in particular, the same maximum. ■

**Algorithm 1.7 (Euclid).** Divide  $a$  by  $b$ , replace  $(a, b) \mapsto (b, r)$ , repeat until the remainder is 0; the  $\gcd$  is **the last nonzero remainder**.

*Justification.* Each step preserves the gcd (Lemma 1.6). The remainders form a strictly decreasing sequence of nonnegative integers ( $0 \leq r < b$ ), which by well-ordering **cannot be infinite**: the algorithm terminates. On reaching  $(d, 0)$ ,  $\gcd(d, 0) = d$ . ■

**Example 1.8.**  $\gcd(252, 198)$ :

$$252 = 1 \cdot 198 + 54 \quad 198 = 3 \cdot 54 + 36 \quad 54 = 1 \cdot 36 + 18 \quad 36 = 2 \cdot 18 + 0$$

$$\gcd(252, 198) = 18.$$

## 1.4 Bézout's identity

**Theorem 1.9 (Bézout).** For  $a, b$  **not both zero**, there exist  $x, y \in \mathbb{Z}$  such that  $\gcd(a, b) = ax + by$ .

*Proof (structural).* Let  $D = \{ax + by : x, y \in \mathbb{Z}\}$  and let  $d$  be the **least positive element** of  $D$  (it exists: since  $a, b$  are not both zero,  $D$  contains some positive element — if  $a \neq 0$ ,  $|a| \in D$ ; if  $a = 0$ , then  $b \neq 0$  and  $|b| \in D$ ; well-ordering). We claim that  $d = \gcd(a, b)$ :

- **$d$  divides  $a$ :** dividing,  $a = dq + r$  with  $0 \leq r < d$ ; then  $r = a - dq$  is a combination of  $a$  and  $b$  (because  $d$  is one), that is,  $r \in D$  with  $0 \leq r < d$ . By minimality of  $d$ ,  $r = 0$ . Likewise for  $b$ .
- **It is the largest:** any common divisor  $c$  of  $a$  and  $b$  divides every combination  $ax + by$  (Prop. 1.2.2), in particular  $d$ ; since  $d > 0$ , this forces  $|c| \leq d$ , and therefore  $c \leq d$ . Thus  $d$  is the largest of the common divisors. ■

**Method 1.10 (extended Euclidean algorithm).** The coefficients  $x, y$  are **constructed** by working back up the divisions of Example 1.8, solving for each remainder:

$$18 = 54 - 36 = 54 - (198 - 3 \cdot 54) = 4 \cdot 54 - 198 = 4(252 - 198) - 198 = 4 \cdot 252 - 5 \cdot 198.$$

$$\text{Check: } 4 \cdot 252 - 5 \cdot 198 = 1008 - 990 = 18 \checkmark.$$

**What Bézout is for (an announcement that will be honored):**

in Session 2 it proves Euclid's lemma (the heart of unique factorization); in Session 3 it **decides and computes inverses** modulo  $n$ ; in Session 4 it builds the solutions of the Chinese remainder theorem. It is, quite possibly, the most profitable result of the module.

## Exercises

1. ★ Compute  $\gcd(1001, 273)$  by the Euclidean algorithm, recording each division, and work back up to write it as  $1001x + 273y$ . Check numerically.
2. ★ Prove from the definition (choose and declare the **type of proof** — M0 connection): if  $a \mid b$  and  $b \mid a$ , then  $a = \pm b$ .

3. ★ Find the quotient and remainder when dividing  $-100$  by  $7$  (mind the convention  $0 \leq r < 7!$ ) and when dividing  $100$  by  $7$ . What is the relation between the two remainders?
4. Justify from the definition:  $\gcd(a, 0) = |a|$  (for  $a \neq 0$ ) and  $\gcd(a, 1) = 1$ .
5. Prove that  $\gcd(a, b) = \gcd(a, b - a)$  **without** using Euclidean division (only Prop. 1.2.2), and explain why this already “almost” justifies the Euclidean algorithm.
6. Give the uniqueness proof of Theorem 1.3 **by contradiction**, and compare it with the direct one in these notes (which do you find clearer? — a connection with M0).
7. Prove: if  $\gcd(a, b) = d$ , then  $\gcd(a/d, b/d) = 1$ . (*Hint: Bézout, divided by  $d$ .*)
8. The set  $D = \{ax + by\}$  from the proof of Bézout: prove that  $D$  is exactly the set of **multiples** of  $d$ . (*The combinations of two numbers are the multiples of their gcd — the fine-grained version of Bézout.*)

## Going further and references

**Going further (gcd of several numbers, and efficiency).**  $\gcd(a, b, c) = \gcd(\gcd(a, b), c)$ , and the Euclidean algorithm is **genuinely fast**: the number of steps is proportional to the number of digits (the worst case is given by the Fibonacci numbers — Lamé’s theorem). That efficiency, set against the cost of factoring, is the crack on which the cryptography of Session 4 is built. See Childs, *A Concrete Introduction to Higher Algebra*, ch. 3–4.

**Going further (Euclidean domains).** Any “world” with a reasonable division with remainder (a *Euclidean domain*) repeats this entire session word for word: gcd, the Euclidean algorithm, Bézout. The star example of the course will be  $K[x]$  (Polynomials module); others, such as the Gaussian integers  $\mathbb{Z}[i]$ , in Aluffi, *Algebra: Chapter 0*, ch. V.

**References.** Childs, ch. 2–4; Dorronsoro–Hernández, *Números, grupos y anillos*, ch. 2.

## Lecture Notes · Arithmetic · Session 2 — Primes and unique factorization

The session’s complete proofs: the infinitude of the primes (with the preliminary lemma the classical proof usually takes for granted), Euclid’s lemma via Bézout, and the fundamental theorem of arithmetic with existence **and uniqueness** in full detail. Exercises and a Going further section at the end.

## 2.1 Prime numbers

**Definition 2.1.** An integer  $p > 1$  is **prime** if its only positive divisors are 1 and  $p$ . An integer  $> 1$  that is not prime is **composite**. (*The number 1 is neither one thing nor the other — the reason for excluding it becomes visible in Theorem 2.6.*)

**Lemma 2.2 (the least divisor is prime).** Every integer  $n > 1$  has some prime divisor; in fact, its **least** divisor greater than 1 is prime.

*Proof.* The set of divisors of  $n$  greater than 1 is nonempty ( $n$  is in it); let  $p$  be its minimum (well-ordering). If  $p$  had a divisor  $d$  with  $1 < d < p$ , that  $d$  would divide  $n$  (transitivity, Prop. 1.2.1) and would be a divisor of  $n$  greater than 1 and smaller than  $p$  — against minimality. Hence  $p$  is prime. ■

**Theorem 2.3 (Euclid: infinitude of the primes).** There are infinitely many prime numbers.

*Proof (by contradiction — the most famous in mathematics).* Suppose there were only finitely many:  $p_1, p_2, \dots, p_k$ . Consider

$$N = p_1 p_2 \cdots p_k + 1.$$

Since  $N > 1$ , it has a prime divisor  $p$  (Lemma 2.2) — which must be one from the list. But each  $p_i$  leaves **remainder** 1 when dividing  $N$  (it divides the product, not the 1; Prop. 1.2.2 with signs). Contradiction:  $p$  is a prime outside the list, which was supposed to be complete. ■

**M0 connection:** this is the canonical example of proof by contradiction (and a good place to recall, in passing, the remark on the excluded middle). Note also the role of Lemma 2.2: the “popular-science” version of this proof usually skips it.

**The proof is constructive (no excluded middle?).** Although presented “by contradiction”, at bottom it **exhibits** a new prime: given any finite list  $p_1, \dots, p_k$ , the number  $N = p_1 \cdots p_k + 1$  has (Lemma 2.2) a prime divisor, and that prime is **not** in the list (each  $p_i$  leaves remainder 1). It is a procedure that, from known primes, produces another — it does not require the excluded middle. The wording “suppose there are finitely many...” is a convenient wrapper, not a logical necessity. (*A nod to the constructivism of M0.*)

## 2.2 Euclid’s lemma — where Bézout pays off

**Lemma 2.4 (general version).** If  $\gcd(c, a) = 1$  and  $c \mid ab$ , then  $c \mid b$ .

*Proof.* Bézout (Theorem 1.9):  $1 = cx + ay$ . Multiplying by  $b$ :

$$b = cbx + aby.$$

$c$  divides the first summand (obvious) and the second (because  $c \mid ab$ ); hence it divides the sum (Prop. 1.2.2):  $c \mid b$ . ■

**Corollary 2.5 (Euclid’s lemma).** If  $p$  is prime and  $p \mid ab$ , then  $p \mid a$  or  $p \mid b$ .

*Proof.* If  $p \nmid a$ , then  $\gcd(p, a) = 1$  (the positive divisors of  $p$  are 1 and  $p$ , and  $p$  does not divide  $a$ ), and Lemma 2.4 gives  $p \mid b$ . ■

**Structural counterfactual (the hypothesis is essential):**  $6 \mid 4 \cdot 9 = 36$ , but  $6 \nmid 4$  and  $6 \nmid 9$ . Composites do not penetrate products: their ability to “split themselves” between the factors ( $6 = 2 \cdot 3$ , with the 2 going to the 4 and the 3 to the 9) is exactly what the lemma forbids the primes. Being prime is not just “not factoring”: it is this property.

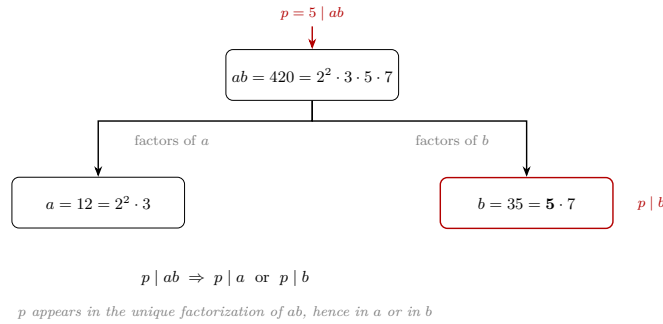


Figure 2: Via unique factorization, the primes of  $ab$  are those of  $a$  together with those of  $b$ ; if  $p \mid ab$ ,  $p$  is one of them, hence it is “in”  $a$  or “in”  $b$ . For a composite like 6, its two primes can split between the factors.

**Heuristic, not proof.** It is a good mental image, but **circular as a proof** (the uniqueness of factorization is proved *with* this lemma): the rigorous argument is Bézout’s (Lemma 2.4).

## 2.3 The fundamental theorem of arithmetic

**Theorem 2.6 (fundamental theorem of arithmetic).** Every integer  $n > 1$  can be written as a product of primes,  $n = p_1 p_2 \cdots p_r$ , and the expression is **unique up to the order** of the factors.

*Proof. Existence (strong induction; see Preliminaries).* For  $n = 2$ : prime, a product of one factor. Let  $n > 2$  and suppose the claim true for all integers between 2 and  $n - 1$ . If  $n$  is prime, done. If not,  $n = ab$  with  $1 < a, b < n$ ; by the induction hypothesis,  $a$  and  $b$  are products of primes, and concatenating, so is  $n$ . (Here the **strong** version of induction is essential:  $a$  and  $b$  are smaller than  $n$  but not necessarily  $n - 1$ , so ordinary induction — which only grants the case  $n - 1$  — would not suffice.)

**Uniqueness.** Suppose  $n = p_1 \cdots p_r = q_1 \cdots q_s$  with all the  $p_i, q_j$  prime; we argue by induction on  $r$ . Since  $p_1 \mid q_1(q_2 \cdots q_s)$ , Euclid’s lemma (applied repeatedly)

forces  $p_1$  to divide **some**  $q_j$ ; reordering,  $p_1 \mid q_1$ , and since both are prime,  $p_1 = q_1$ . Cancelling  $(n/p_1)$ :  $p_2 \cdots p_r = q_2 \cdots q_s$ , a **smaller** number, to which the induction hypothesis applies:  $r - 1 = s - 1$  and the factors coincide up to order. (Base case  $r = 1$ :  $n = p_1$  prime admits no factorizations other than itself.) ■

**Remark 2.7 (why 1 is not prime).** If 1 counted as a prime, uniqueness would die on the spot:  $6 = 2 \cdot 3 = 1 \cdot 2 \cdot 3 = 1 \cdot 1 \cdot 2 \cdot 3 = \cdots$ . The exclusion is not a whim of nomenclature: it is the condition for Theorem 2.6 to be true as stated.

**Remark 2.7b (negative integers and “units”).** Theorem 2.6 is stated for  $n > 1$  with positive primes, and then “up to order” is exact. If negative integers are allowed, uniqueness becomes “up to order **and a sign**  $\pm 1$ ”: the  $\pm 1$  are the **units** of  $\mathbb{Z}$  (its invertible elements), and the general statement of unique factorization — in any unique factorization domain — is “up to order and units”. Here the positive case suffices; the term “unit” is formalized in ring theory (references in the Going further section).

*Structural lens:*  $\mathbb{Z}$  has **atoms** (the primes) and unique composition. The complete skeleton — division with remainder  $\rightarrow$  the Euclidean algorithm  $\rightarrow$  Bézout  $\rightarrow$  Euclid’s lemma  $\rightarrow$  unique factorization — will be rebuilt, piece by piece and with the same proofs, in  $K[x]$  (Polynomials module). When it arrives, it will not be a new topic: it will be this one, in disguise.

## 2.4 Practical consequences

**Proposition 2.8 (gcd and lcm via exponents).** For nonzero  $a, b$ , writing  $|a| = \prod p_i^{\alpha_i}$  and  $|b| = \prod p_i^{\beta_i}$  (exponents  $\geq 0$ , common primes), then

$$\gcd(a, b) = \prod p_i^{\min(\alpha_i, \beta_i)}, \quad \text{lcm}(a, b) = \prod p_i^{\max(\alpha_i, \beta_i)}, \quad \gcd \cdot \text{lcm} = |a b|.$$

(The sign of  $a, b$  is irrelevant for divisibility — hence the  $|\cdot|$  and the  $|ab|$ , not  $ab$ , valid also with negatives. Proof: a divisor of  $a$  has, by the uniqueness in the fundamental theorem of arithmetic, exponents  $\leq \alpha_i$ ; the rest is taking minima/maxima and the identity  $\min + \max = \alpha + \beta$ . Details: exercise 5.)

**But beware the practical trap:** this formula requires **factoring**, and factoring is computationally expensive; the Euclidean algorithm computes the gcd **without factoring**, and blazingly fast. That asymmetry — multiplication/gcd cheap, factoring expensive — is not an anecdote: it is the basis of the cryptography of Session 4.

**Proposition 2.9 (primality test up to  $\sqrt{n}$ ).** If  $n > 1$  has no prime divisor  $p \leq \sqrt{n}$ , then  $n$  is prime.

*Proof.* If  $n = ab$  with  $1 < a \leq b < n$ , then  $a^2 \leq ab = n$ , so  $a \leq \sqrt{n}$ , and the least prime divisor of  $a$  (Lemma 2.2) is a prime divisor of  $n$  less than or equal to  $\sqrt{n}$ . ■

## Exercises

1. ★ Factor 360 and 756; compute the gcd and lcm via exponents, and **redo the gcd with the Euclidean algorithm**. Compare the work of the two methods and explain which would scale better with 100-digit numbers.
2. ★ (Counterfactual.) Find **your own** counterexample to Euclid’s lemma with a **composite number** (different from  $6 \mid 4 \cdot 9$ ) and point out how the composite “splits” between the factors.
3. ★ Is 221 prime? And 223? Decide with the test of Proposition 2.9, listing exactly which primes must be tried and why those suffice.
4. ★ Answer with the fundamental theorem of arithmetic: why is 1 not prime? (The correct answer is not “because the definition says so”.)
5. Complete the proof of Proposition 2.8 (divisors via exponents; the min + max identity).
6. Prove that  $\sqrt{2}$  is irrational using the fundamental theorem of arithmetic: in  $a^2 = 2b^2$ , count the parity of the exponent of 2 on each side. (*Compare with the “classical” proof by contradiction from M0 — same conclusion, finer tool.*)
7. Prove that there are infinitely many primes of the form  $4k + 3$ . (*Guide: adapt Euclid’s proof with  $N = 4p_1 \cdots p_k - 1$ ; you will need: a product of numbers of the form  $4k + 1$  is of the form  $4k + 1$ .*)
8. If  $p$  is prime and  $p \mid a^n$ , prove that  $p^n \mid a^n$ . (*Iterated Euclid’s lemma + fundamental theorem of arithmetic.*)

## Going further and references

**Going further (the distribution of the primes).** Knowing there are infinitely many opens the question of *how many*: the **prime number theorem** says that up to  $x$  there are approximately  $x/\ln x$ . Between the arithmetic of this session and that result lie a century and a half of analysis. A readable overview: Childs, or ch. 1 of Hardy–Wright, *An Introduction to the Theory of Numbers*.

**Going further (unique factorization that fails).** There are “worlds” where the fundamental theorem of arithmetic is false: in  $\mathbb{Z}[\sqrt{-5}]$ ,  $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$  are two genuinely different factorizations. Uniqueness is not free: it depends on Euclid’s lemma, which depends on Bézout, which depends on division with remainder. Aluffi, ch. V.

**References.** Childs, ch. 5–6; Dorronsoro–Hernández, ch. 2–3.

## Lecture Notes · Arithmetic · Session 3 — Congruences and $\mathbb{Z}/n\mathbb{Z}$

The heart of the module, with everything proved: the equivalence relation, the *well-definedness* of the quotient operations, the invertibility criterion (with the effective computation of inverses), the **zero divisor / non-invertible dichotomy** — which prepares the notion of a **field**, to be formalized later in the course —,  $\mathbb{Z}/p\mathbb{Z}$  as a first field, and Fermat's little theorem with its rearrangement proof. Exercises and a Going further section at the end.

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### 3.1 Congruences

**Definition 3.1.** Fix  $n \geq 2$ :  $a \equiv b \pmod{n}$  if  $n \mid (a - b)$ .

**Proposition 3.2.**  $a \equiv b \pmod{n} \iff a$  and  $b$  leave the **same remainder** when divided by  $n$ .

*Proof.* Writing  $a = nq_1 + r_1$ ,  $b = nq_2 + r_2$  with  $0 \leq r_i < n$  (Theorem 1.3):  $a - b = n(q_1 - q_2) + (r_1 - r_2)$ . If  $r_1 = r_2$ , then  $n \mid (a - b)$ . Conversely, if  $n \mid (a - b)$ , then  $n \mid (r_1 - r_2)$  (Prop. 1.2.2), and since  $|r_1 - r_2| < n$ , it must be that  $r_1 = r_2$  (the bounding trick from Session 1). ■

**Proposition 3.3.** Congruence modulo  $n$  is an **equivalence relation** on  $\mathbb{Z}$  (the concept from M0, at work!).

*Proof. Reflexive:*  $n \mid 0$ . *Symmetric:* if  $n \mid (a - b)$  then  $n \mid (b - a)$  (the negative). *Transitive:* if  $n \mid (a - b)$  and  $n \mid (b - c)$ , then  $n \mid ((a - b) + (b - c)) = (a - c)$  (Prop. 1.2.2). ■

**Definition 3.4.** The **class** of  $a$  is  $\bar{a} = \{b : b \equiv a \pmod{n}\} = \{a, a \pm n, a \pm 2n, \dots\}$ . By Proposition 3.2 there are exactly  $n$  **classes**, one per remainder:  $\bar{0}, \bar{1}, \dots, \overline{n-1}$ .

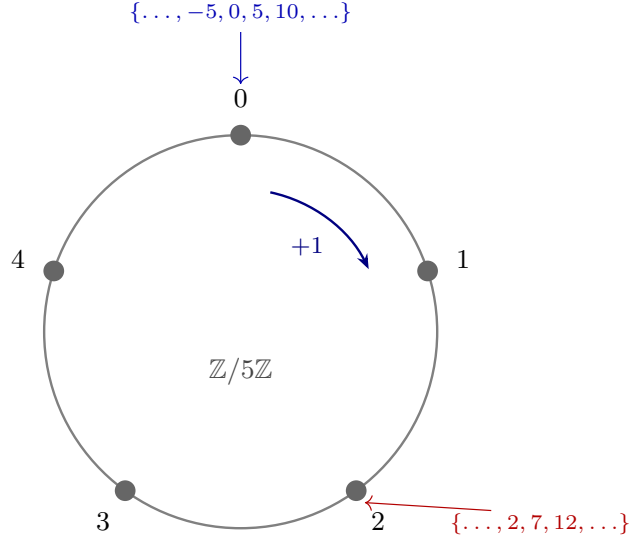


Figure 3: Congruence modulo  $n$ : wrap the number line around a circle with  $n$  marks; integers congruent modulo  $n$  land on the same mark (here  $n = 5$ ).

### 3.2 $\mathbb{Z}/n\mathbb{Z}$ : operating with classes (and the toll of the quotient)

**Definition 3.5.**  $\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$ , with

$$\bar{a} + \bar{b} := \overline{a + b}, \quad \bar{a} \cdot \bar{b} := \overline{a \cdot b}.$$

**Proposition 3.6 (the operations are well defined).** If  $a \equiv a' \pmod{n}$  and  $b \equiv b' \pmod{n}$ , then  $a + b \equiv a' + b'$  and  $ab \equiv a'b' \pmod{n}$ .

*Proof.* Sum:  $(a + b) - (a' + b') = (a - a') + (b - b')$ , a sum of two multiples of  $n$ . Product: the trick of **adding and subtracting the cross term**:

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b'),$$

a combination of multiples of  $n$  (Prop. 1.2.2). ■

**Why this is THE fine point:** Definition 3.5 computes with **representatives** of the classes. If the result depended on the chosen representative, the definition would be nonsense. Proposition 3.6 is the **mandatory toll of every quotient construction** — a pattern worth internalizing here, in the most concrete example possible (M0: equivalence classes).

The properties of the operations (associative, commutative, distributive, identity elements  $\bar{0}, \bar{1}$ , negatives) are **inherited** from  $\mathbb{Z}$  representative by representative. *Structural lens*: a set with these properties is called a **ring** — we drop the name in passing; its general theory is not part of this course.

### 3.3 Invertibles and zero divisors

**Definition 3.7.**  $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$  is **invertible** if there exists  $\bar{x}$  with  $\bar{a}\bar{x} = \bar{1}$ . And  $\bar{a} \neq \bar{0}$  is a **zero divisor** if there exists  $\bar{b} \neq \bar{0}$  with  $\bar{a}\bar{b} = \bar{0}$ .

**Theorem 3.8 (invertibility criterion).**  $\bar{a}$  is invertible in  $\mathbb{Z}/n\mathbb{Z} \iff \gcd(a, n) = 1$ . Moreover, in that case the inverse is **computed** with the extended Euclidean algorithm.

*Proof.* ( $\Leftarrow$ ) Bézout:  $1 = ax + ny$  for some  $x, y$ . Modulo  $n$ :  $\bar{a}\bar{x} = \bar{1}$ , that is,  $\bar{a}\bar{x} = \bar{1}$  — and the Bézout coefficients **are** the inverse. ( $\Rightarrow$ ) If  $\bar{a}\bar{x} = \bar{1}$ , then  $n \mid (ax - 1)$ , i.e.  $ax - 1 = ny$ :  $1 = ax - ny$ . Every common divisor of  $a$  and  $n$  divides that combination (Prop. 1.2.2), hence divides 1: the gcd is 1. ■

**Example 3.9.** Inverse of  $\bar{7}$  in  $\mathbb{Z}/30\mathbb{Z}$ : the Euclidean algorithm gives  $30 = 4 \cdot 7 + 2$ ,  $7 = 3 \cdot 2 + 1$ ; working back up,  $1 = 7 - 3 \cdot 2 = 7 - 3(30 - 4 \cdot 7) = 13 \cdot 7 - 3 \cdot 30$ . Hence  $\bar{7}^{-1} = \bar{13}$ . *Check*:  $7 \cdot 13 = 91 = 3 \cdot 30 + 1$  ✓.

**Theorem 3.10 (the dichotomy).** Let  $\bar{a} \neq \bar{0}$  in  $\mathbb{Z}/n\mathbb{Z}$ . Then:

$$\bar{a} \text{ is a zero divisor} \iff \bar{a} \text{ is not invertible} \iff \gcd(a, n) > 1.$$

*Proof.* The second equivalence is Theorem 3.8 negated. We prove the first in both directions.

(Zero divisor  $\Rightarrow$  not invertible.) Let  $\bar{a}\bar{b} = \bar{0}$  with  $\bar{b} \neq \bar{0}$ . If  $\bar{a}$  had an inverse, multiplying:  $\bar{b} = (\bar{a}^{-1}\bar{a})\bar{b} = \bar{a}^{-1}(\bar{a}\bar{b}) = \bar{a}^{-1}\bar{0} = \bar{0}$  — contradiction.

(Not invertible  $\Rightarrow$  zero divisor.) Let  $d = \gcd(a, n) > 1$  and take  $\bar{b} = \overline{n/d}$ . Then  $\bar{b} \neq \bar{0}$  (because  $0 < n/d < n$ , since  $d > 1$ ), and

$$a \cdot \frac{n}{d} = \frac{a}{d} \cdot n \equiv 0 \pmod{n}$$

(it is an integer multiple of  $n$ :  $a/d \in \mathbb{Z}$  because  $d \mid a$ ). Hence  $\bar{a}\bar{b} = \bar{0}$ : a zero divisor. ■

**Remark 3.11 (the dichotomy is special to  $\mathbb{Z}/n\mathbb{Z}$ ).** In  $\mathbb{Z}$ , the number 2 is **not** invertible and is **not** a zero divisor either: outside the  $\mathbb{Z}/n\mathbb{Z}$ 's the two notions need not fill the whole space. (*This is exactly the caution that will reappear later, when the **fields** are presented in Linear Algebra: here it is already proved in the most concrete example.*)

**Corollary 3.12 (the thread of fields begins).** 1. If  $p$  is **prime**, every nonzero element of  $\mathbb{Z}/p\mathbb{Z}$  is invertible — a finite world where **one can divide**. It gets a *name*:  $\mathbb{Z}/p\mathbb{Z}$  is a **field** (the formal definition and its theory, in the

Linear Algebra module). 2. If  $n$  is **composite**,  $\mathbb{Z}/n\mathbb{Z}$  has zero divisors and is **not** a field.

*Proof.* (1)  $0 < a < p$  and  $p$  prime  $\Rightarrow \gcd(a, p) = 1$  (the only divisors of  $p$  are 1 and  $p$ )  $\Rightarrow$  invertible (3.8). (2)  $n = ab$  with  $1 < a, b < n$  gives  $\bar{a}\bar{b} = \bar{0}$  with both nonzero. ■

| $\times$ | 0 | 1 | 2 | 3 |
|----------|---|---|---|---|
| 0        | 0 | 0 | 0 | 0 |
| 1        | 0 | 1 | 2 | 3 |
| 2        | 0 | 2 | 0 | 2 |
| 3        | 0 | 3 | 2 | 1 |

$\mathbb{Z}/4\mathbb{Z}$ : there are zero divisors  
( $2 \cdot 2 = 0$ ,  $2 \neq 0$ )

| $\times$ | 0 | 1 | 2 | 3 | 4 |
|----------|---|---|---|---|---|
| 0        | 0 | 0 | 0 | 0 | 0 |
| 1        | 0 | 1 | 2 | 3 | 4 |
| 2        | 0 | 2 | 4 | 1 | 3 |
| 3        | 0 | 3 | 1 | 4 | 2 |
| 4        | 0 | 4 | 3 | 2 | 1 |

$\mathbb{Z}/5\mathbb{Z}$ : a field  
(no product of nonzero elements is 0)

Figure 4: Multiplication tables of  $\mathbb{Z}/4\mathbb{Z}$  and  $\mathbb{Z}/5\mathbb{Z}$ : in  $\mathbb{Z}/4\mathbb{Z}$  there are zero divisors ( $2 \cdot 2 = 0$ );  $\mathbb{Z}/5\mathbb{Z}$ , since 5 is prime, is a field.

### 3.4 Fermat's little theorem

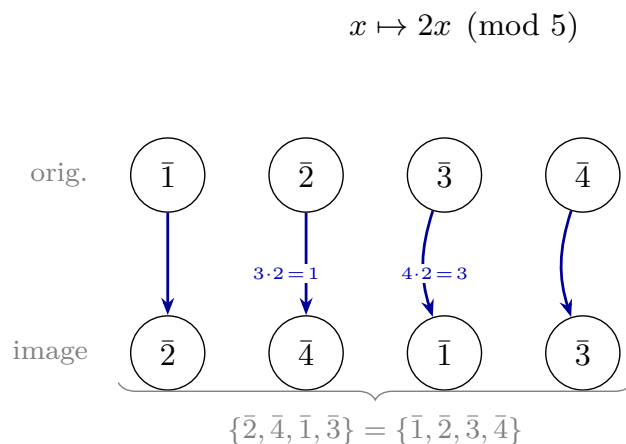
**Theorem 3.13 (Fermat).** If  $p$  is prime and  $p \nmid a$ , then

$$a^{p-1} \equiv 1 \pmod{p}.$$

*Proof (the rearrangement).* Consider the nonzero classes  $\bar{1}, \bar{2}, \dots, \overline{p-1}$  and multiply them all by  $\bar{a}$ :

$$\bar{a} \cdot \bar{1}, \bar{a} \cdot \bar{2}, \dots, \bar{a} \cdot \overline{p-1}.$$

**Claim: this list is the same as before, rearranged.** Indeed, the map  $\bar{x} \mapsto \bar{a}\bar{x}$  is **injective** ( $\bar{a}\bar{x} = \bar{a}\bar{y} \Rightarrow \bar{x} = \bar{y}$ , multiplying by  $\bar{a}^{-1}$ , which exists by Corollary 3.12.1) and never produces zero ( $\bar{a}\bar{x} = \bar{0}$  with  $\bar{a}$  invertible would force  $\bar{x} = \bar{0}$ ); an injective map from a **finite** set to itself is bijective (M0). Equating the products of the two lists:



*the same set, shuffled*

Figure 5: For  $p = 5$ ,  $a = 2$ : the row  $(\bar{1}, \bar{2}, \bar{3}, \bar{4})$  is transformed into  $(\bar{2}, \bar{4}, \bar{1}, \bar{3})$  by  $x \mapsto 2x$  ( $1 \rightarrow 2$ ,  $2 \rightarrow 4$ ,  $3 \rightarrow 1$ ,  $4 \rightarrow 3$ ) — the same set, shuffled.

That is why the product of the whole row does not change, and that is what gets cancelled.

$$\overline{(p-1)!} = \overline{a^{p-1}} \cdot \overline{(p-1)!}.$$

And  $\overline{(p-1)!}$  is invertible (a product of invertibles); cancelling it:  $\overline{a^{p-1}} = \bar{1}$ . ■

Note what the proof uses: that  $\mathbb{Z}/p\mathbb{Z}$  is a **field** (all nonzero elements invertible, twice) and a **counting** argument (injective on a finite set = bijective, M0). **What it is for:** Fermat powers the modular exponentiation that sustains public-key cryptography — Session 4 collects the debt.

## Exercises

- ★ In  $\mathbb{Z}/30\mathbb{Z}$ , compute  $\overline{17}^{-1}$  with the extended algorithm (not by trial and error) and check.
- ★ Classify **all** the nonzero elements of  $\mathbb{Z}/12\mathbb{Z}$  into invertibles and zero divisors; verify the dichotomy of Theorem 3.10 and exhibit, for each zero divisor, the partner  $\bar{b}$  that betrays it.
- ★ Compute the remainder of  $3^{100}$  when divided by 7 (Fermat:  $3^6 \equiv 1$ ) and that of  $2^{1000}$  when divided by 11.

4. ★ **(The Linear Algebra counterfactual, now with theory.)** In  $\mathbb{Z}/6\mathbb{Z}$ :  $2 \cdot 3 = \bar{0}$ . Locate the gcd that explains it via Theorem 3.10, and construct the “canonical” zero divisor  $n/\bar{d}$  for  $\bar{a} = 2$ .
5. Prove that  $\mathbb{Z}$  has no zero divisors, and conclude that the dichotomy 3.10 **fails** in  $\mathbb{Z}$  (with 2 as the witness).
6. Prove the **divisibility rule for 9**: a number is divisible by 9 if and only if the sum of its digits is. (*Hint*:  $10 \equiv 1 \pmod{9}$ , so  $10^k \equiv 1$ .)
7. Is it true that  $a \equiv b \pmod{n}$  implies  $a^2 \equiv b^2 \pmod{n}$ ? And the converse? Prove or refute (minimal counterexample).
8. **(SageMath, asynchronous activity.)** Generate the tables of  $\mathbb{Z}/5\mathbb{Z}$  and  $\mathbb{Z}/6\mathbb{Z}$  (`Integers(n)`), spot the zero divisors visually, compute inverses with `inverse_mod`, and check Fermat with `power_mod(a, p-1, p)` for several primes.

## Going further and references

**Optional going further (rings).** A set with two operations  $+, \cdot$  satisfying the properties inherited here (both associative, addition commutative, distributive, with identity elements  $\bar{0}, \bar{1}$  and negatives) is a **commutative ring with identity**;  $\mathbb{Z}$ ,  $\mathbb{Z}/n\mathbb{Z}$  and (Polynomials module)  $K[x]$  are examples. The **zero divisors** and the **units** (invertibles) of this session are general notions of ring theory; a commutative ring without zero divisors is an **integral domain**, and one where every nonzero element is invertible is a **field** (M4). To expand, optional: Dorronsoro–Hernández, *Números, grupos y anillos*, ch. 3; Childs, ch. 7–9; with a categorical flavor, Aluffi, *Algebra: Chapter 0*, ch. III. Not assessed.

**Going further (Euler’s  $\varphi$  function).** How many invertibles does  $\mathbb{Z}/n\mathbb{Z}$  have? Exactly  $\varphi(n) = \#\{1 \leq a \leq n : \gcd(a, n) = 1\}$ . **Euler’s** theorem generalizes Fermat’s:  $a^{\varphi(n)} \equiv 1 \pmod{n}$  if  $\gcd(a, n) = 1$  — and the same rearrangement proof works, restricted to the invertibles. For  $n = pq$  (the RSA case):  $\varphi(pq) = (p-1)(q-1)$ . Childs, ch. 9; Dorronsoro–Hernández.

**References.** Childs, ch. 7–9 (congruences,  $\mathbb{Z}/n\mathbb{Z}$ , Fermat); Dorronsoro–Hernández, ch. 3. The dichotomy and fields: Linear Algebra notes, Session 1 (where what is proved here gets used).

## Lecture Notes · Arithmetic · Session 4 — The Chinese remainder theorem and cryptography (RSA)

The module’s closing session with everything proved: the constructive CRT (with the uniqueness lemma), fast modular exponentiation with

the why behind it, and the theorem that **RSA decrypts** — Fermat and the CRT glued together, with the toy example verified by hand and the code of the demo. Exercises and a Going further section at the end.

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## 4.1 The Chinese remainder theorem

**Theorem 4.1 (CRT).** If  $\gcd(m, n) = 1$ , the system of congruences

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}$$

has a solution, and it is **unique modulo  $mn$**  (any two solutions are congruent modulo  $mn$ ).

*Proof.* **Existence (constructive: the switches).** Bézout:  $1 = mx_0 + ny_0$ . Define

$$e_1 = ny_0, \quad e_2 = mx_0.$$

Then  $e_1 + e_2 = 1$ , and:  $e_1 \equiv 1 \pmod{m}$  (since  $e_1 = 1 - mx_0$ ) and  $e_1 \equiv 0 \pmod{n}$ ; symmetrically  $e_2 \equiv 0 \pmod{m}$ ,  $e_2 \equiv 1 \pmod{n}$ . They are **switches**: each is worth 1 in one modulus and 0 in the other. The combination

$$x = a e_1 + b e_2$$

satisfies  $x \equiv a \cdot 1 + b \cdot 0 = a \pmod{m}$  and  $x \equiv b \pmod{n}$ . ■ (*existence*)

|                                                                                                              | $m = 3$ | $n = 5$ |
|--------------------------------------------------------------------------------------------------------------|---------|---------|
| $e_1 = 10$                                                                                                   | 1 on    | 0 OFF   |
| $e_2 = 6$                                                                                                    | 0 OFF   | 1 on    |
| $e_1 \equiv 1 \pmod{3}, \quad e_1 \equiv 0 \pmod{5}$<br>$e_2 \equiv 0 \pmod{3}, \quad e_2 \equiv 1 \pmod{5}$ |         |         |
| $x \equiv a \pmod{3}, \quad x \equiv b \pmod{5}$<br>$x = a e_1 + b e_2 = 10a + 6b \pmod{15}$                 |         |         |

Figure 6: The CRT “switches”:  $e_1$  is worth 1 modulo  $m$  and 0 modulo  $n$ , and  $e_2$  the other way around; combining them,  $x = a e_1 + b e_2$  solves the system.

**Lemma 4.2 (for the uniqueness).** If  $m \mid x$ ,  $n \mid x$  and  $\gcd(m, n) = 1$ , then  $mn \mid x$ .

*Proof.*  $x = mt$ ; since  $n \mid mt$  and  $\gcd(n, m) = 1$ , Lemma 2.4 gives  $n \mid t$ , i.e.  $t = ns$ :  $x = mns$ . ■

**Uniqueness of the CRT.** If  $x$  and  $x'$  solve the system,  $m$  and  $n$  divide  $x - x'$ , and by Lemma 4.2,  $mn \mid (x - x')$ . ■

**Corollary 4.2b (CRT for  $k$  moduli).** If  $m_1, \dots, m_k$  are **pairwise coprime**, the system  $x \equiv a_i \pmod{m_i}$  ( $i = 1, \dots, k$ ) has a solution, unique modulo  $m_1 m_2 \cdots m_k$ .

*Proof.* Induction on  $k$ . The case  $k = 2$  is Theorem 4.1. For the step,  $M = m_1 \cdots m_{k-1}$  is coprime with  $m_k$  (a prime of  $m_k$  divides no  $m_i$ ,  $i < k$ , by pairwise coprimality; fundamental theorem of arithmetic): solve the first  $k-1$  congruences by hypothesis (one congruence modulo  $M$ ) and combine it with the  $k$ -th one via the two-moduli case. (*Switch version:*  $e_i \equiv 1 \pmod{m_i}$  and  $\equiv 0$  in the others, built with  $\widehat{m}_i = \prod_{j \neq i} m_j$ , which is invertible modulo  $m_i$ ;  $x = \sum_i a_i e_i$ . Exercise 5.) ■

**Example 4.3 (the riddle).**  $x \equiv 2 \pmod{3}$ ,  $x \equiv 3 \pmod{5}$ . Bézout for  $(3, 5)$ :  $1 = 2 \cdot 3 - 1 \cdot 5$ , so  $e_1 = -5 \pmod{15}$  ( $\equiv 1 \pmod{3}$ ,  $\equiv 0 \pmod{5}$ ) and  $e_2 = 6$ . Solution:  $x = 2(-5) + 3(6) = 8$ , unique modulo 15:  $x \in \{8, 23, 38, \dots\}$ . Check:  $8 = 2 + 2 \cdot 3$  ✓,  $8 = 3 + 5$  ✓.

**Non-coprime moduli (the hypothesis is not decorative).** With  $\gcd(m, n) > 1$  the system has a solution **if and only if**  $\gcd(m, n) \mid (b - a)$  (exercise 6); when there is one, it is unique modulo  $\text{lcm}(m, n)$ .

- *Unsolvable:*  $x \equiv 0 \pmod{2}$ ,  $x \equiv 1 \pmod{4}$ :  $\gcd(2, 4) = 2 \nmid (1 - 0) = 1$  — no solution ( $x$  even and at the same time  $\equiv 1 \pmod{4}$ , impossible).
- *Solvable despite non-coprimality:*  $x \equiv 2 \pmod{4}$ ,  $x \equiv 4 \pmod{6}$ :  $\gcd(4, 6) = 2 \mid (4 - 2) = 2$  — there is a solution, unique modulo  $\text{lcm}(4, 6) = 12$ . Trying:  $x = 10$  satisfies  $10 \equiv 2 \pmod{4}$  and  $10 \equiv 4 \pmod{6}$ ; the solutions are  $x \equiv 10 \pmod{12}$ .

## 4.2 Fast modular exponentiation

For  $a^k \pmod{n}$  with  $k$  huge, the key is to **square repeatedly**, reducing modulo  $n$  at every step, and multiply the squares dictated by the **binary expansion** of  $k$ :

$$k = 13 = 8 + 4 + 1 = 1101_2 \implies a^{13} = a^8 \cdot a^4 \cdot a^1.$$

**Example 4.4.**  $3^{13} \pmod{7}$ :  $3^2 = 9 \equiv 2$ ;  $3^4 \equiv 2^2 = 4$ ;  $3^8 \equiv 4^2 = 16 \equiv 2$ . Then  $3^{13} = 3^8 \cdot 3^4 \cdot 3 \equiv 2 \cdot 4 \cdot 3 = 24 \equiv 3 \pmod{7}$ .

**Cost:** on the order of the **number of bits of  $k$**  — about  $2 \log_2 k$  multiplications, and  $\log_2 k$  is precisely that number of bits —: for a thousand-bit  $k$ , **a few thousand** operations, not  $2^{1000}$ . Without this algorithm, RSA would be theory without practice.

### 4.3 RSA

#### What it solves: public-key cryptography

Before the scheme, the problem. To **encrypt** a message is to transform it with a key so that only whoever holds the right key can recover it. *Classical* (symmetric) cryptography uses the **same** secret key to encrypt and to decrypt: sender and receiver must **share it beforehand** over a secure channel — but if they already had a secure channel, a good part of the original problem would already be solved. That vicious circle — the **key-exchange problem** — limited cryptography for centuries.

The idea of **public-key** cryptography breaks it. Each user has a mathematically linked **pair** of keys: a **public** one, published openly, which anyone uses to encrypt messages for them, and a **private** one, kept secret, which is the only one capable of decrypting. No secret ever needs to be shared in advance. **RSA** (Rivest–Shamir–Adleman, 1978) was the first practical system to realize this idea, and it still protects — among many other things — the web connections with a “padlock”. What keeps the public key from betraying the private one is a **computational asymmetry** (multiplying is easy, factoring is not), which we make precise in “*Why it is secure*”.

#### The scheme

1. **Keys.** Choose primes  $p \neq q$  (large);  $n = pq$ ;  $\varphi = (p-1)(q-1)$ ; choose  $e$  with  $\gcd(e, \varphi) = 1$ , and compute  $d = e^{-1} \bmod \varphi$  — **the extended algorithm from S1/S3, once again**. **Public key:**  $(n, e)$ . **Private key:**  $d$ .
2. **Encrypt** (anyone):  $c = m^e \bmod n$ . **Decrypt** (only whoever holds  $d$ ):  $m = c^d \bmod n$ . (Messages  $0 \leq m < n$ .)

**Theorem 4.5 (RSA decrypts).** With the keys built this way, for every  $m$ :  $(m^e)^d \equiv m \pmod{n}$ .

*Proof.* Since  $ed \equiv 1 \pmod{\varphi}$ , we write  $ed = 1 + k(p-1)(q-1)$ . **We work modulo  $p$  and modulo  $q$  separately, and glue with the CRT.**

*Modulo  $p$ :* if  $p \mid m$ , both sides are  $\equiv 0$ . If  $p \nmid m$ , Fermat (Theorem 3.13) gives  $m^{p-1} \equiv 1$ , so

$$m^{ed} = m \cdot (m^{p-1})^{k(q-1)} \equiv m \cdot 1 = m \pmod{p}.$$

*Modulo  $q$ :* identical. Thus  $p$  and  $q$  both divide  $m^{ed} - m$ ; since  $\gcd(p, q) = 1$ , **Lemma 4.2** gives  $pq \mid (m^{ed} - m)$ , that is,  $m^{ed} \equiv m \pmod{n}$ . ■

Inventory of the proof: **Bézout** (built  $d$ ), **Fermat** (the engine), **CRT/Lemma 4.2** (the glue). The entire module, in a ten-line proof.

### Why it is secure (honesty)

To decrypt one needs  $d$ ; to compute  $d$  one needs  $\varphi = (p-1)(q-1)$ ; and knowing  $\varphi$  together with  $n$  is **equivalent to factoring**  $n$  (exercise 8: from  $n$  and  $\varphi$  one recovers  $p$  and  $q$  by solving a quadratic). Security rests, then, on the **computational asymmetry** of Session 2: multiplying is instantaneous; factoring a 2048-bit  $n$ , infeasible with known technology.

**Multiplying versus factoring (the asymmetry, with intuition).** It is worth seeing why one direction is easy and the other is not. **Multiplying** two large numbers is fast: the work grows *smoothly* with the number of **digits** — the school algorithm uses on the order of (number of digits)<sup>2</sup> operations; one says it is **polynomial** in the size of the input (the “size” is the number of digits, not the value of the number). For the reverse path — given  $n$ , **finding its divisors** — no fast method is known: in essence one must **search** among candidates, and that search space grows **explosively** (exponentially) with the number of digits. The gap becomes abysmal as soon as the numbers are large: with two primes of about 300 digits each ( $\approx 1024$  bits), multiplying them is instantaneous, while factoring the  $\sim 600$ -digit product would exceed the age of the universe using all of today’s computers at once. *That* distance — trivial in one direction, intractable in the other — is the security of RSA. And it is robust against hardware progress: if the machines improve, **enlarging the keys** makes encryption slightly more expensive and factoring vastly more so.

**Note (quantum computing).** “Infeasible” refers to **classical** computers. In 1994 Peter Shor discovered an algorithm that factors **efficiently**... on a **quantum** computer. That RSA still stands today only means that quantum machines large enough to factor real keys do not yet exist. That is why **post-quantum cryptography** — based on problems conjectured to be hard even for quantum computers — is already being standardized. (Going further; cf. Childs.)

*(Honest fine print: “equivalent de facto” — real-world RSA also carries random padding and implementation precautions without which there are attacks that do not require factoring. Ours is the mathematical core.)*

### Example 4.6 (a toy one, verified by hand)

$p = 3, q = 11$ :  $n = 33, \varphi = 20$ . We take  $e = 3$  ( $\gcd(3, 20) = 1$ ); the extended algorithm gives  $d = 7$  ( $3 \cdot 7 = 21 \equiv 1 \pmod{20}$ ). **Encrypt**  $m = 5$ :  $c = 5^3 = 125 \equiv 26 \pmod{33}$ . **Decrypt**:  $26^7 \pmod{33}$ , by squaring ( $26 \equiv -7$ ):  $(-7)^2 = 49 \equiv 16$ ;  $(-7)^4 \equiv 16^2 = 256 \equiv 25$ ;  $26^7 \equiv 25 \cdot 16 \cdot (-7) \equiv 4 \cdot (-7) = -28 \equiv 5 \pmod{33}$   
✓ — the message comes back.

### The SageMath demo (skeleton)

```
p = next_prime(10^6); q = next_prime(2*10^6)
n = p*q; phi = (p-1)*(q-1)
e = 65537; assert gcd(e, phi) == 1
d = inverse_mod(e, phi)
m = 271828                                     # the "message"
c = power_mod(m, e, n)                         # encrypt (public key)
assert power_mod(c, d, n) == m                 # decrypt (private key) - m comes back
```

(Seed of the SE3 mini-project: each student generates their own keys and messages are exchanged; guided asynchronous activity.)

---

### Exercises

1. ★ Solve with the switches:  $x \equiv 3 \pmod{7}$ ,  $x \equiv 5 \pmod{11}$ . Give the general solution and the least positive one; check.
2. ★ RSA by hand with  $p = 5$ ,  $q = 11$ ,  $e = 3$ : build  $n, \varphi, d$ ; encrypt  $m = 4$  and decrypt the result, all by fast exponentiation.
3. ★ Compute  $7^{45} \bmod 11$  combining **Fermat** (reduce the exponent) and successive squaring.
4. ★ (SageMath — SE3 seed.) Set up the full scheme with primes of 6–8 digits; encrypt a numeric message and verify the decryption. Then **break** someone else's toy RSA with **factor(n)** and discuss: up to what size of  $n$  does **factor** succeed on your machine within a second?
5. Complete the **switch construction** of Corollary 4.2b: for pairwise coprime  $m_1, \dots, m_k$ , define  $\widehat{m}_i = \prod_{j \neq i} m_j$ , find its inverse modulo  $m_i$ , form the  $e_i$  and verify that  $x = \sum_i a_i e_i$  solves the system. Apply it to  $x \equiv 2 \pmod{3}$ ,  $x \equiv 3 \pmod{5}$ ,  $x \equiv 2 \pmod{7}$ .
6. (Counterfactual completed.) Prove:  $x \equiv a \pmod{m}$ ,  $x \equiv b \pmod{n}$  has a solution **if and only if**  $\gcd(m, n) \mid (b - a)$ . (This explains exactly when the non-coprime case fails.)
7. Why does RSA require  $\gcd(e, \varphi) = 1$ ? Point out the step of the construction that would break without it.
8. Prove that knowing  $n = pq$  and  $\varphi = (p - 1)(q - 1)$  allows one to recover  $p$  and  $q$ . (Hint:  $p + q = n - \varphi + 1$ ; then  $p, q$  are the roots of  $t^2 - (p + q)t + n$ .)

### Going further and references

**Going further (from Fermat to Euler, and general RSA).** Our Theorem 4.5 avoids the general  $\varphi$  function by using Fermat + CRT. The alternative route: **Euler's** theorem ( $a^{\varphi(n)} \equiv 1$  if  $\gcd(a, n) = 1$ ; Going further of Session 3) gives the case  $\gcd(m, n) = 1$  in one line — but it does not cover the messages not coprime with  $n$ , which our proof does. Childs, ch. 9–10.

**Going further (digital signatures).** Swapping the roles of the keys ( $d$  to “encrypt”,  $e$  to verify), RSA produces **signatures**: only the owner of  $d$  could have generated the signature, and anyone can verify it with the public key. The same mathematics, read backwards. Childs, ch. 10.

**References.** CRT: Childs, ch. 8; Dorronsoro–Hernández. RSA: Childs, ch. 10 (the reference closest to the course); the original Rivest–Shamir–Adleman paper (1978) is surprisingly readable.